

ERDEM KUS

AI Researcher & Engineer · GenAI, Quantum Machine Learning and Optimisation

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EXECUTIVE PROFILE

AI research and engineering specialist with 6+ years of industry experience translating advanced methods into secure, scalable production systems. Architected core RAG, agentic AI, retrieval, inference and observability capabilities for Nokia's global enterprise GenAI platform, serving **1,000+ users across 1M+ documents**. Combines deep research in cost-aware active learning, algorithm selection and quantum optimisation with hands-on engineering across LLM orchestration, AI infrastructure and enterprise security. **Trusted technical authority** who converts complex domain needs into production-ready architectures and guides multidisciplinary engineering teams through implementation.

SELECTED IMPACT

1M+

enterprise
documents

1,000+

Nokia support
engineers

2,300+

researchers
supported

90%

lower annotation
cost

175+

CPU-years of
experiments

CORE VALUE PROPOSITION

- **Enterprise GenAI architecture:** production RAG, permission-aware retrieval, multi-agent workflows, model routing, AI observability and GPU-backed inference.
- **Research-to-production execution:** evaluates emerging methods rigorously, selects fit-for-purpose approaches and turns them into reliable solutions for real users.
- **Technical influence without title dependency:** owns critical architecture decisions, mentors engineers and leaders, and aligns domain, support, backend, frontend and platform teams.

PROFESSIONAL EXPERIENCE

Research Fellow – AI for Academic Synergy

Nov 2025 – Present

University of St Andrews · St Andrews, UK

- Developing a GenAI recommendation platform for **2,300+ researchers**, creating structured personas from academic profiles and identifying high-potential collaborations within St Andrews and across UK universities.
- Built LLM-based extraction and synthesis pipelines over structured and unstructured Pure data to identify research areas, methods and keywords; extended matching to open funding opportunities.
- Implemented **evidence-level provenance** linking recommendations to supporting papers, profiles and datasets, strengthening transparency, traceability and auditability.
- Contributing to a **£3M Wellcome Trust-funded** programme spanning the Universities of St Andrews, Edinburgh and Glasgow.

GenAI Engineer & Quantum Machine Learning Specialist

Apr 2020 – Aug 2026

Nokia · Istanbul, Türkiye / Global

- **Architected NICO**, Nokia's production enterprise GenAI/RAG platform, consolidating fragmented technical knowledge into natural-language search, question answering, summarisation, reasoning and tool-enabled workflows for **1,000+ global daily users across 1M+ documents**.
- **Owned key architecture decisions** across the RAG pipeline, document ingestion, parsing, chunking, embeddings and vector indexing; separated backend and RAG/LLM services and established Celery/Redis asynchronous processing with Qdrant-based retrieval.
- **Designed and developed the agentic AI architecture from the ground up** using LangGraph, MCP and a dynamic Tool Registry, including planner, tool selector, executor, reviewer/reviser and final-answer agents.
- Enabled production enterprise workflows spanning Jira/Confluence search, deep research, Excel calculation and extraction, and presentation generation through context- and permission-aware tool selection.
- Designed **permission-aware retrieval** with subgroup definitions, collection-level restrictions and retrieval-time access checks, adding a critical control layer against cross-group data leakage.
- Evaluated Qwen, Llama, Mixtral and GPT-family models against quality, latency and deployment requirements; introduced **LLM-based reranking** to strengthen context selection at 1M+ document scale.
- Defined GPU/Triton inference architecture, KV caching and inference optimisation approaches; used caching and ReWOO planning to reduce redundant LLM calls and repeated context processing across agentic workflows.
- Established AI observability through MLflow, Grafana and production monitoring; deployed across multiple environments and guided engineering implementation across backend, frontend, DevOps/platform, domain and support teams.
- Acted as a **de facto technical lead and go-to expert** for GenAI, RAG, agentic AI, LLM architecture and optimisation; delivered

global technical presentations, knowledge-sharing sessions and mentoring to engineers and managers.

SELECTED DEEP-TECH INITIATIVE · D-WAVE QUANTUM TRANSFORMATION

- Identified and evaluated Nokia use cases suitable for quantum transformation, assessing problem structure and quantum suitability before translating selected mobile fingerprinting and network topology optimisation problems into QUBO/Ising formulations in direct collaboration with D-Wave.
- Benchmarked quantum approaches against classical methods, identifying stronger results in selected limited-data scenarios and moving the end-to-end solution beyond research/PoC into production.

Backend / Artificial Intelligence Engineer

Oct 2019 – Mar 2020

Cliexa · Istanbul, Türkiye

- Developed Python REST APIs for real-time ingestion of vital-sign data from patient wearables, supporting continuous health monitoring.
- Built and deployed predictive machine-learning models on patient assay data to support clinical decision-making and treatment planning; developed a Xamarin application for structured patient-data collection.

TECHNICAL EXPERTISE

- **GenAI & Agentic Systems:** RAG, LLM agents, multi-agent systems, LangGraph, LangChain, LlamaIndex, MCP, ReWOO, GraphRAG, tool routing, prompt and context orchestration.
- **Retrieval & Enterprise AI:** Qdrant, FAISS, semantic search, embeddings, chunking, vector indexing, LLM-based reranking, permission-aware retrieval, data isolation.
- **AI Infrastructure & MLOps:** FastAPI, Flask, Celery, Redis, PostgreSQL, MariaDB, Triton Inference Server, GPU inference, KV caching, MLflow, Grafana, Prometheus.
- **Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, active learning, uncertainty quantification, algorithm selection, cost-aware ML.
- **Quantum Machine Learning & Optimisation:** D-Wave Ocean and Leap, Qiskit, PennyLane, QUBO, Ising models, quantum annealing, combinatorial optimisation.
- **Cloud & High-Performance Computing:** Azure, AWS, Vertex AI, Slurm, Cirrus, MedicineBow, Hypatia, large-scale experimentation.
- **Programming:** Python, Java, JavaScript, SQL, R, Perl.

EDUCATION

PhD in Computer Science

Sep 2022 – Aug 2026

University of St Andrews · St Andrews, UK

- **Thesis:** Frugal Algorithm Selection: Cost-Aware Active Learning for Efficient Training Data Collection in Algorithm Selection.
- Addressed the high and non-uniform cost of collecting algorithm-performance labels, where runtime observations are often right-censored by timeouts and epistemic uncertainty-only acquisition can become biased toward timed-out instances.
- Designed surrogate-based, cost-aware and multi-objective acquisition strategies, achieving up to **90%** lower labelling cost while maintaining competitive algorithm-selection performance.
- Evaluated 28 strategies across 16 ASLib scenarios through large-scale Slurm experiments on Cirrus, MedicineBow and Hypatia, totalling approximately **175+ CPU-years** of computation.

MSc in Artificial Intelligence – GPA 4.00/4.00

2020 – 2022

Bahcesehir University · Istanbul, Türkiye

- **Thesis:** Hybrid Fairness for Fair Decision-Making. Academic Excellence /
- Outstanding Achievement Scholarship

RESEARCH IMPACT & SELECTED PUBLICATIONS

SUBMITTED MANUSCRIPT

- [1] **HiCAL: Hierarchical Cost-Aware Active Learning for Frugal Algorithm Selection**
Kus, E., Akgun, O., Dang, N., Kotthoff, L., & Miguel, I. · AAAI 2027 · Under review

PEER-REVIEWED PUBLICATIONS

- [2] **Frugal Algorithm Selection for Combinatorial Search**
Kus, E., Kotthoff, L., Akgun, O., Dang, N., & Miguel, I. · *Journal of Artificial Intelligence Research (JAIR)*, 2026 · DOI
- [3] **On the Effect of Training Data Selection in Automated Algorithm Selection**
Kus, E., Kotthoff, L., Akgun, O., Dang, N., & Miguel, I. · *Principles and Practice of Constraint Programming (CP 2026)*, 2026 · DOI
- [4] **Frugal Algorithm Selection**
Kus, E., Akgun, O., Dang, N., & Miguel, I. · *Principles and Practice of Constraint Programming (CP 2024)*, 2024 · DOI
- [5] **Cost-Efficient Training for Automated Algorithm Selection**
Kus, E., Akgun, O., Dang, N., & Miguel, I. · *AutoML 2024 Workshop*, 2024 · OpenReview

SELECTED TALKS

- **Invited Seminar**, Okinawa Institute of Science and Technology (OIST), Japan, Jul 2026: *Frugal Algorithm Selection for Combinatorial Search and its Adaptation to Frugal Training of Noise-Aware Quantum Neural Networks*.
- **Conference Talk**, FLoC / CP 2026, Lisbon, Portugal, Jul 2026: *On the Effect of Training Data Selection in Automated Algorithm Selection*.
- **PGR Seminar**, School of Computer Science, University of St Andrews, St Andrews, Scotland, Oct 2025: *Frugal Algorithm Selection for Combinatorial Search*.
- **CIRCA Seminar**, Centre for Interdisciplinary Research in Computational Algebra, University of St Andrews, St Andrews, Scotland, Nov 2024: *Frugal Algorithm Selection*.
- **Conference Talk**, CP 2024, Girona, Spain, Sep 2024: *Frugal Algorithm Selection*.
- **Workshop Poster**, AutoML 2024, Paris, France, Sep 2024: *Cost-Efficient Training for Automated Algorithm Selection*.
- **Invited Talk**, NoBel AI Community, Nokia Bell Labs, Antwerp / Virtual, Jan 2023: *Quantum Machine Learning, QUBO Formulations and Hamiltonian-Based Optimisation*.
- **Invited Talk**, NoBel AI Community, Nokia Bell Labs, Antwerp / Virtual, Jul 2021: *Fair AI and Hybrid Group/Individual Fairness*.

TECHNICAL LEADERSHIP, TEACHING & RECOGNITION

Graduate Teaching Assistant – Lead Lab Demonstrator & Tutor

Feb 2023 – Jun 2026

University of St Andrews · St Andrews, UK

- Led undergraduate and MSc programming laboratories and tutorials covering Python, Java, object-oriented programming, programming with data, code quality and software-engineering practice.
- Provided structured assessment and feedback across coursework and project work; translated complex technical concepts into practical learning for varied experience levels.

GLOBAL MENTORING & KNOWLEDGE SHARING

- Mentor in Nokia global programmes for managers and engineers; contributed to gender-equality mentoring and AI knowledge-sharing initiatives.
- Invited speaker for Nokia Bell Labs' NoBel AI Community on quantum optimisation, QUBO, Qiskit and fair AI; delivered global sessions on RAG, agentic AI and architecture.
- Invited seminar speaker at Okinawa Institute of Science and Technology (2026); conference presenter at FLoC / CP 2026, CP 2024 and AutoML 2024.

ACADEMIC SERVICE & COMMUNITY

- **Conference Reviewer**: AutoML Conference (2026).
- **Journal Reviewer**: Information Sciences, Elsevier (2025–present); IEEE Transactions on Artificial Intelligence (2024–present); Engineering Applications of Artificial Intelligence, Elsevier (2023–present).
- **Volunteer AI Engineer, Humanitrack and Memorix (2020)**: developed NLP knowledge-hub assessment and ML-based personalised learning solutions.